Introduction to Software Engineering

Project Proposal & Plan

The student team is required to complete the **Project Proposal & Plan** documentation for the assigned course project, following the attached template.



Software Engineering Department
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Project Proposal & Plan

Objectives

This document focus on the following topics:

- ✓ Completing the Project Proposal & Plan document with the following sections:
  - Vision
  - Proposed Solution
  - Project Plan
  - Risk Management
  - Tools & Collaboration Setup
- ✓ Including an AI usage log for this phase
- ✓ Understanding the Project Proposal & Plan document.

1 Member Contribution Assessment

ID	Name	Role(s)	Contribution (%)	Evidence
24127378	Nguyễn Huy Hoàng	PM / Planning	25%	WBS, Estimation, Phase Gates, Roles – GitHub Issue/PR link: https://github.com/gloryh241106/ShiftManager/pull/1
24127085	Võ Quang Minh	BA / Architect	25%	C4 Diagram, Technology Stack, Tools Setup – GitHub Issue/PR link: https://github.com/gloryh241106/ShiftManager/pull/2
24127499	Lê Hoàng Phúc	BA / Reporting	25%	Problem Statement, Product Position, MoSCoW – GitHub Issue/PR link
24127568	Nguyễn Cao Trí	Scheduling / Risk	25%	Gantt Chart, Cost Estimate, Risk Register – GitHub Issue/PR link

2 Vision

2.1 Preliminary Problem Statement

Business Context and Problem Overview

ShiftManager targets the shift scheduling and coordination problems of a single production team within a factory in an industrial park. In the proposed scenario, managers use Excel to prepare work schedules and Zalo groups to communicate shifts and last-minute changes. As the number of workers increases, maintaining a consistent, timely and fair schedule becomes increasingly difficult.

The project focuses on four operational pain points:

1. Scattered schedule information: Employees must search through past chat messages to find their latest assigned shifts. Spreadsheet versions and chat announcements may become inconsistent.

2. Uncommunicated shift changes: Last-minute changes may not reach every affected employee, and managers cannot easily determine who has seen an updated schedule.

3. Unexpected absences and staffing shortages: Employees may report an absence late or fail to attend an assigned shift. Without an integrated coverage workflow, managers spend time identifying available replacements.

4. Manual scheduling and perceived unfairness: Assigning many workers to shifts using spreadsheets can cause overlaps, understaffed shifts and uneven allocation of work hours or night shifts.

ShiftManager proposes a centralized web application that generates a draft roster using explicit scheduling constraints, allows managers to review and publish schedules, gives employees access to an authoritative roster, and supports absence reporting and replacement coordination. A limited Boss dashboard provides an overview of staffing coverage and workload distribution.

This is a proposed factory-team scenario discussed during project planning. The team has not yet conducted an on-site industrial survey, so the pain points are treated as assumptions to be validated rather than measured findings.

Operating Environment

- Client-Side Environment: ShiftManager is a responsive web application accessible via standard modern web browsers supporting HTML5, CSS3, and JavaScript across desktop and mobile devices. The frontend user interface is built with React (JavaScript).
- Server-Side Environment: The application backend is powered by Node.js and the Express framework, exposing a RESTful API with role-based access control.
- Database and Cloud Infrastructure: Persistent relational data and transactional operations are hosted on Supabase (PostgreSQL relational database). Source code version control, pull-request reviews, and automated Continuous Integration (CI) pipelines are managed through Git/GitHub and GitHub Actions.

Design and Implementation Constraints

- Role-Based Access & Pre-created Accounts: The system restricts capabilities across three application roles (Employee, Manager, and Boss) using pre-created, seeded demo accounts and fictional business data.
- Scheduling and Absence Management Business Rules:
    - Employees must meet the required skills and availability conditions before being assigned to a shift.
    - Shift assignments must not overlap and must respect the configured working-hour and rest-time constraints.
    - Automatic scheduling generates a draft roster; the Manager must review and publish it before it becomes official.
    - If no eligible employee is available, the system must report an uncovered shift rather than create an invalid assignment.
    - A replacement assignment requires the substitute employee's acceptance and the Manager's confirmation.
- Data Consistency: A confirmed replacement assignment must update the employee assignment and coverage status consistently within a database transaction.
- Privacy & Visibility Boundaries: Employees can view their own roster and relevant schedule changes. Managers can manage their production team's schedules, while Boss users can access authorized staffing and workload summaries.
- Schedule & Process Constraints: Development must adhere to the course's waterfall phase gates, Conventional Commits standards (feat:, fix:, docs:, test:), protected main branch rules requiring at least one peer review per pull request.

2.2 Product Position Statement

Element	Content
For	Factory shift managers, production supervisors, and employees working in a single production team.
Who	Experience fragmented schedule communication, unexpected staffing shortages, and difficulty creating balanced shift assignments using spreadsheets and chat groups.
The (product name)	ShiftManager, a lightweight web-based factory shift scheduling and coordination system.
That	Generates constraint-based draft schedules, centralizes published rosters, tracks schedule acknowledgements, and assists managers in covering unexpected absences.
Unlike	Disconnected Excel and Zalo workflows or larger workforce-management platforms that include payroll, complex attendance systems, and enterprise-wide HR functions.
Our product	Focuses on one production team with explainable scheduling criteria, visible roster changes, and a simple absence-to-coverage workflow.

2.3 Stakeholders and Users

#	Stakeholder / user	Description	Main interests
1	**Factory owner / Production supervisor (Boss)**	Reviews team-wide staffing coverage and workload distribution.	Operational visibility, fair allocation, fewer uncovered shifts.
2	**Shift manager (Manager)**	Configures staffing requirements, generates and publishes rosters, and handles absence coverage.	Efficient planning, fewer conflicts, adequate staffing.
3	**Production worker (Employee)**	Views assigned shifts, acknowledges updates, and reports absences.	Clear schedules, timely updates, predictable assignments.
4	**Lecturer / TA (Academic stakeholder)**	Evaluates the project deliverables and demonstration; not an application user.	Correct documentation and working prototype.

2.4 Scope and Key Features

The ShiftManager MVP consists of four Must-have features (F01–F04) designed for a single production team within a factory. The system focuses on role-based access, constraint-based shift scheduling, centralized schedule communication, and absence reporting with shift coverage. Features outside this scope are excluded from the PA5 implementation baseline.

ID	Feature	Description	Priority (M/S/C/W)
F01	Authentication & Role-Based Access	Authenticate users with Supabase Auth and restrict access based on Employee, Manager, and Boss roles.	Must
F02	Constraint-Based Automatic Shift Scheduling	Generate a proposed weekly roster based on staffing demand, employee eligibility, availability, non-overlapping shifts, and workload-balancing rules. Allow managers to review and publish the roster.	Must
F03	Centralized Schedule & Change Acknowledgement	Provide an official roster, highlight schedule changes, allow employees to acknowledge that they have viewed updates, and provide managers and bosses with appropriate schedule visibility	Must
F04	Absence Reporting & Shift Coverage	Allow employees to report inability to attend assigned shifts. Managers can record unreported absences, identify eligible replacement candidates, obtain replacement acceptance, and update the roster.	Must

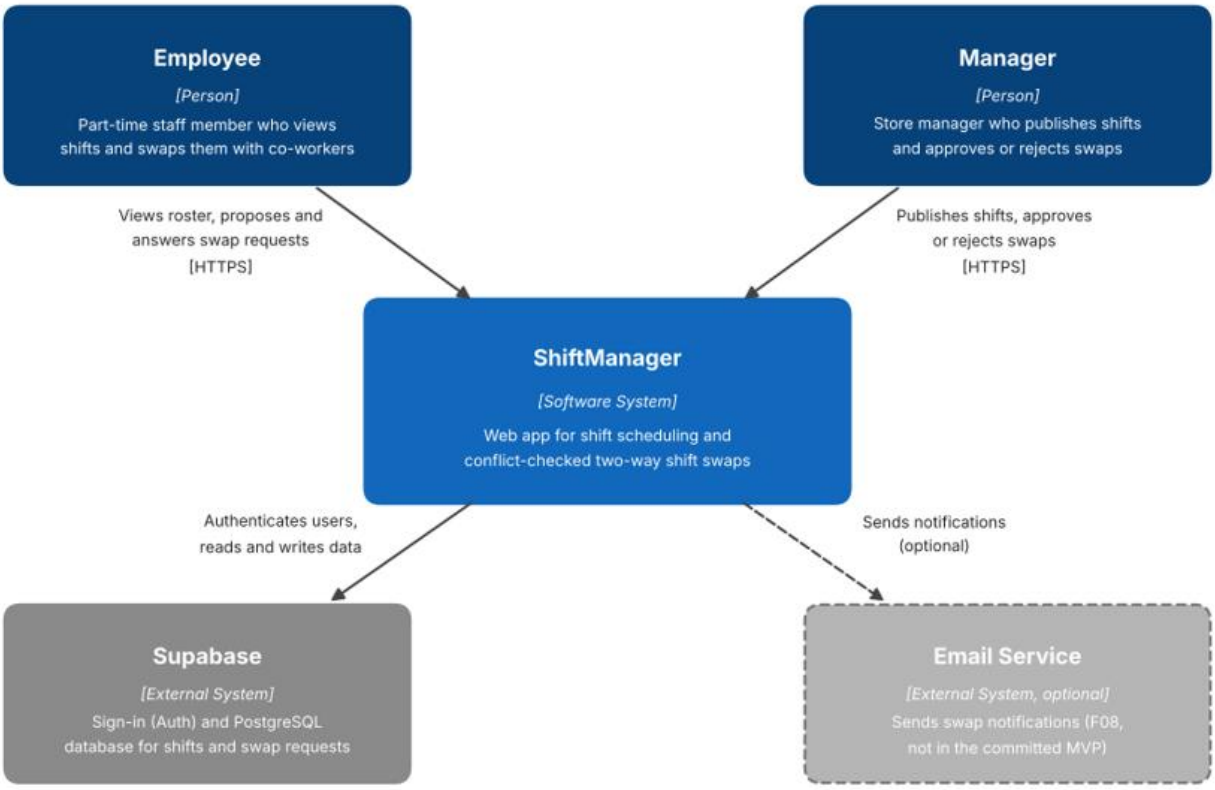
Out of scope:

- ✓ Time Tracking & Attendance: Employee clock-in/clock-out functionality and timesheet tracking.
- ✓ Payroll & Compensation: Wage calculation, overtime pay processing, or financial reporting.
- ✓ Full Leave Management: Vacation requests, sick leave tracking, or complex recurring availability forms.
- ✓ Uncontrolled Shift Transfers: Employees cannot transfer shifts directly without an approved coverage workflow.

- ✓ Multi-Department Operations: Managing multiple production departments, factories, or cross-department workforce transfers.
- ✓ Push Notifications & External HR Integrations: Mobile push notifications and integrations with third-party HR or accounting systems.

3 Proposed Solution

3.1 Preliminary Architecture



The preliminary C4 System Context diagram represents the initial project concept. Following the TA consultation on 19 October 2026, the system scope was revised. The architecture diagram will be updated in subsequent PAs to reflect the new scope and user roles, as permitted by the TA.

3.2 Technology Stack

Layer	Technology	Reason for choice
Front end	React	Popular and easy to learn; builds the web interface from reusable components.
Back end	Node.js (Express)	Uses JavaScript like the front end and is simple for building the REST API.
Database	Supabase (PostgreSQL)	Free, easy to set up, and provides a relational database with built-in sign-in.
Hosting / deployment	Vercel	Free and easy to deploy from GitHub, so the team can show a live demo.
Testing & CI	GitHub Actions	Automatically runs tests on every pull request.

3.3 Hardware and Operating Environment

ShiftManager is a web application, so it needs no special hardware.

Environment	Hardware / devices	Software and network
Development	Laptops or desktops of team members, with internet access.	Node.js, Git, a code editor and a modern browser.
Testing	Same laptops, plus one smartphone for mobile checks.	Chrome, Edge or Firefox; automated tests run on GitHub Actions.
Production / live demo	No dedicated hardware; cloud services only.	Deployed on Vercel; database on Supabase.
End-user client	Any computer, tablet or phone with internet.	A modern web browser; nothing to install.

4 Project Plan

4.1 Process and Phase Gates

The project follows the waterfall model. Each phase ends with a phase gate (a PA submission). After a PA is submitted it becomes a baseline; later changes must be updated in the document and listed in the next PA.

PA	Phase	Due	Exit criteria
PA1	Initiation & planning	End of week 3	Reviewed vision, stakeholders and scope; at least 3-level WBS; person-hour estimates with assumptions; Gantt/dependencies/critical path; at least 5 risks with RE and owners; working collaboration tools.
PA2	Requirements	End of week 5	Reviewed SRS with numbered/prioritized FRs F01–F04 covering authentication, automatic scheduling, schedule acknowledgement, and absence/coverage management, measurable NFRs, business rules and acceptance criteria; use case model and specifications; key mockups; requirements traceability and review log.
PA3	Design	End of week 7	Reviewed design architecture (C4 and deployment), at least 3 ADRs, class/sequence/state diagrams, normalized relational ERD with table specs/indexes, screen specifications and requirements/design consistency review.
PA4	Test planning	End of week 8	Reviewed test plan (levels, types, environment, entry/exit criteria, defect workflow), EP/BVA, decision tables, repeatable test cases and traceability for all Must-have FRs.
PA5	Implementation, testing & deployment	End of week 10	Deployed and demonstrable F01–F04 with constraint-based schedule generation, published roster acknowledgement, and absence coverage workflow; test execution and defect records; unit tests with coverage; automation of at least 2 use cases × at least 2 scenarios; CI on pull requests, code reviews, demo video and retrospective.
Final	Presentation & demo	Week 11	Complete slides and live demonstration; each of the four members participates and can explain the system, including at least 2 minutes of speaking per member as stated in the rubric.

Following the TA consultation on 19 October 2026, the project scope was revised from small-business two-way shift swapping to factory-team shift scheduling and absence coverage. The previous F01–F06 scope is superseded by four Must-have features (F01–F04). The WBS, estimates, schedule, architecture, risk register and requirements baseline must be updated accordingly.

4.2 Work Breakdown Structure (WBS)

The WBS is deliverable-oriented and has four nested levels: project (0.0), phase/workstream (1.0–7.0), deliverable group (x.y), and assignable work package (x.y.z). The following IDs match Section 4.3 and must be used consistently in the team task board and Gantt schedule. This revised WBS covers the four Must-have features F01–F04 for a single factory production team. The previous two-way shift swap workflow has been removed from the implementation baseline. Work packages reflect automatic scheduling, centralized roster acknowledgement, and absence coverage.

1.0 Project Initiation & Planning (PA1)

- ***1.1 Vision & Scope***
  - *1.1.1 Preliminary Problem Statement*
  - *1.1.2 Product Position Statement*
  - *1.1.3 MoSCoW priorities, scope & exclusions*
  - *1.1.4 Stakeholders and user interests*
- ***1.2 Preliminary Solution***
  - *1.2.1 C4 System Context diagram*
  - *1.2.2 Technology stack and operating environment*
- ***1.3 Project Planning***
  - *1.3.1 Phase gates, exit criteria, roles*
  - *1.3.2 Three-level WBS for whole project*
  - *1.3.3 Estimate work-package person-hours*
  - *1.3.4 Gantt schedule, dependencies & critical path*
  - *1.3.5 Human resources and project cost*
- ***1.4 Risk & Tools***
  - *1.4.1 Risk register, RE, mitigation and owners*
  - *1.4.2 Repo, board, comms and access evidence*
- ***1.5 PA Submission***
  - *1.5.1 Final review, contribution, AI log, export*

2.0 Requirements Engineering (PA2)

- ***2.1 Software Requirements***
  - *2.1.1 Functional requirements F01–F04 (FRs)*
  - *2.1.2 NFRs, business rules & acceptance criteria*
- ***2.2 Use Case Modeling***
  - *2.2.1 Actors, Use Case diagram*
  - *2.2.2 Use Case specifications incl exceptions*
- ***2.3 Requirements Validation***

2.3.1 UI mockups and screen flows
2.3.2 Traceability and review issue log
2.4 PA Submission
2.4.1 SRS final review, contribution, AI log
3.0 Software Design (PA3)
3.1 Software Architecture
3.1.1 C4 Container, Component, Deployment
3.1.2 At least 3 ADRs with justification
3.2 Object-Oriented Design
3.2.1 Class diagram & class specs
3.2.2 Sequence & State diagrams
3.3 Database Design
3.3.1 ERD and normalized schema
3.3.2 Constraints, table specs & indexes
3.4 UI/UX Design
3.4.1 Screen specs, flow, accessibility checklist
3.5 PA Submission
3.5.1 Design/SRS traceability review & finalize PA3
4.0 Test Planning (PA4)
4.1 Test Strategy
4.1.1 Test Plan, environment, entry/exit, defect process
4.2 Test Design
4.2.1 Equivalence Partitioning and BVA design
4.2.2 Decision tables for scheduling and absence/coverage rules
4.3 Test Cases & Coverage
4.3.1 Detailed test cases for F01–F04
4.3.2 Requirements–test traceability review
4.4 PA Submission
4.4.1 Final test plan review, contribution & AI log
5.0 Implementation, Testing & Deployment (PA5)
5.1 Development Foundation
5.1.1 Create React/Express project and local dev config
5.1.2 Supabase schema, migrations, employee profiles, skills, availability, shift data and test data
5.2 Authentication & Shift Management
5.2.1 Supabase Auth, JWT verification and Employee/Manager/Boss authorization (F01)
5.2.2 Constraint-based automatic scheduling and fairness scoring (F02)
5.2.3 Published roster, change indicators, acknowledgement and Boss overview (F03)
5.2.4 Manager roster preview, manual adjustment and publishing (F02)
5.3 Absence Reporting & Shift Coverage
5.3.1 Employee absence submission and manager no-show recording (F04)
5.3.2 Replacement eligibility checks and substitute acceptance (F04)
5.3.3 Manager coverage decision and consistent roster update (F04)
5.3.4 Integrate absence/coverage workflow UI with REST API (F04)

5.4 Software Verification
5.4.1 Unit tests and coverage report
5.4.2 Automated cases (2 use cases × 2 scenarios), PR CI
5.4.3 Test execution, defect fixes and PR reviews
5.5 Deployment & Delivery
5.5.1 Deploy and verify live demo
5.5.2 Demo video, implementation report, retrospective
6.0 Final Presentation & Demo
6.1 Presentation Preparation
6.1.1 Prepare slides and project summary
6.1.2 Demo script + practice all speakers
6.2 Final Delivery
6.2.1 Final demo and presentation
7.0 Project Management & Monitoring
7.1 Weekly Monitoring
7.1.1 Weekly status meetings W02–W10
7.1.2 Weekly Status Report W03
7.1.3 Weekly reports W02 and W04–W10 (8 reports)
7.2 Project Control
7.2.1 Maintain GitHub task board
7.2.2 Git/PR reviews, risks, baselines, change records

4.3 Estimation

WBS ID	Work package	Owner	Effort (person-hours)	Method / assumptions
1.1.1	Preliminary Problem Statement	Phúc	3	Hypothetical business case; no real business survey is claimed.
1.1.2	Product Position Statement	Phúc	2	Documentation and review of PA1 deliverable
1.1.3	MoSCoW priorities, scope & exclusions	Phúc	3	Keep F01–F04 Must-have;
1.1.4	Stakeholders and user interests	Minh	1.5	Separate stakeholder analysis to provide a single accountable owner.
1.2.1	C4 System Context diagram	Minh	3	Only the C4 System Context diagram is required in PA1.
1.2.2	Technology stack and operating environment	Minh	2	React + Node.js/Express + Supabase Auth + Supabase Database (PostgreSQL) stack

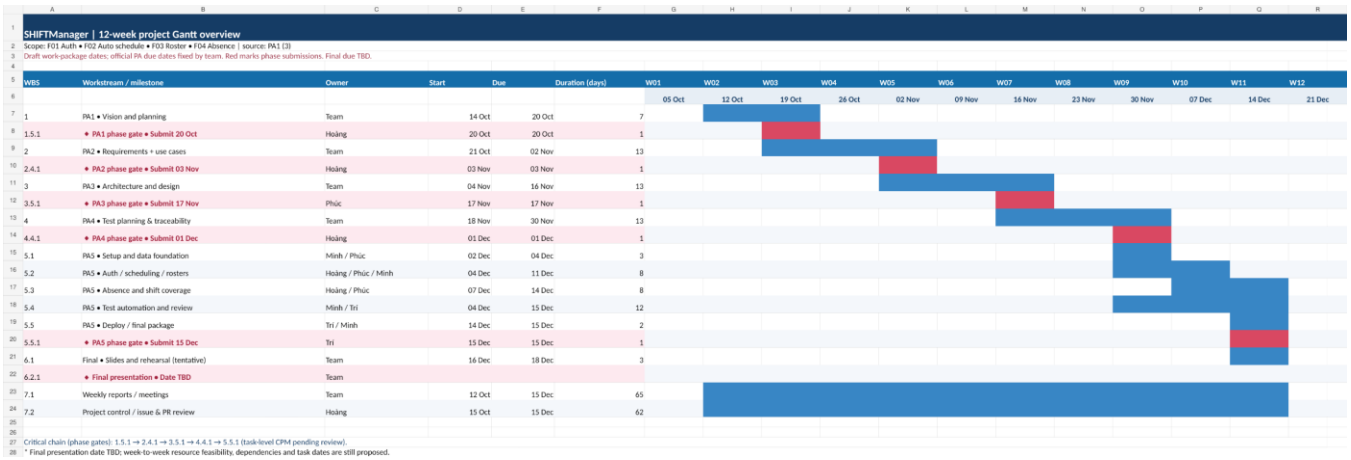
1.3.1	Phase gates, exit criteria, roles	Hoàng	1.5	Documentation and review of PA1 deliverable
1.3.2	Three-level WBS for whole project	Hoàng	3.5	Hand the initial WBS draft to Trí as early as possible.
1.3.3	Estimate work-package person-hours	Hoàng	3	Initial estimate; review and revise with the team.
1.3.4	Gantt schedule, dependencies & critical path	Trí	4	Identify the critical path only after checking durations and dependencies.
1.3.5	Human resources and project cost	Trí	2	Hourly rate is an assumption; Trí must propose and justify it.
1.4.1	Risk register, RE, mitigation and owners	Trí	4	>= 5 risks; RE theo thang trong lecture.
1.4.2	Repo, board, comms and access evidence	Minh	2.5	Create a TA-accessible channel; protect main and review PRs.
1.5.1	Final review, contribution, AI log, export	Hoàng	2	Collect actual contribution evidence and actual AI usage from all members.
2.1.1	Functional requirements F01–F04 (FRs)	Phúc	6	Specification and peer review of PA2 deliverable
2.1.2	NFRs, business rules & acceptance criteria	Trí	7	Specification and peer review of PA2 deliverable
2.2.1	Actors, Use Case diagram	Phúc	4	Specification and peer review of PA2 deliverable
2.2.2	Use Case specifications incl exceptions	Hoàng	10	Specification and peer review of PA2 deliverable
2.3.1	UI mockups and screen flows	Minh	10	Specification and peer review of PA2 deliverable
2.3.2	Traceability and review issue log	Trí	5	Specification and peer review of PA2 deliverable
2.4.1	SRS final review, contribution, AI log	Hoàng	4	Specification and peer review of PA2 deliverable
3.1.1	C4 Container, Component, Deployment	Minh	7	Design and documentation of PA3 deliverable
3.1.2	At least 3 ADRs with justification	Hoàng	5	Design and documentation of PA3 deliverable

3.2.1	Class diagram & class specs	Phúc	7	Design and documentation of PA3 deliverable
3.2.2	Sequence & State diagrams	Hoàng	10	Design and documentation of PA3 deliverable
3.3.1	ERD and normalized schema	Trí	7	Design and documentation of PA3 deliverable
3.3.2	Constraints, table specs & indexes	Trí	5	Design and documentation of PA3 deliverable
3.4.1	Screen specs, flow, accessibility checklist	Minh	6	Design and documentation of PA3 deliverable
3.5.1	Design/SRS traceability review & finalize PA3	Phúc	4	Design and documentation of PA3 deliverable
4.1.1	Test Plan, environment, entry/exit, defect process	Phúc	8	Test design and review of PA4 deliverable
4.2.1	Equivalence Partitioning and BVA design	Hoàng	8	Test design and review of PA4 deliverable
4.2.2	Decision tables for scheduling and absence/coverage rules	Minh	6	Test design and review of PA4 deliverable
4.3.1	Detailed test cases for F01–F04	Trí	11	Test design and review of PA4 deliverable
4.3.2	Requirements–test traceability review	Minh	4	Test design and review of PA4 deliverable
4.4.1	Final test plan review, contribution & AI log	Hoàng	3	Test design and review of PA4 deliverable
5.1.1	Create React/Express project and local dev config	Minh	7	Implementation, debugging and peer review
5.1.2	Supabase schema, migrations, employee profiles, skills, availability, shift data and test data	Phúc	7	Implementation, debugging and peer review
5.2.1	Supabase Auth, JWT verification and Employee/Manager/Boss authorization (F01)	Hoàng	10	Implementation, debugging and peer review
5.2.2	Constraint-based automatic scheduling and fairness scoring (F02)	Phúc	10	Implementation, debugging and peer review
5.2.3	Published roster, change indicators, acknowledgement and Boss overview (F03)	Minh	8	Implementation, debugging and peer review
5.2.4	Manager roster preview, manual adjustment and publishing (F02)	Hoàng	8	Implementation, debugging and peer review

5.3.1	Employee absence submission and manager no-show recording (F04)	Hoàng	8	Implementation, debugging and peer review
5.3.2	Replacement eligibility checks and substitute acceptance (F04)	Phúc	7	Implementation, debugging and peer review
5.3.3	Manager coverage decision and consistent roster update (F04)	Hoàng	13	High risk: Transaction and data consistency.
5.3.4	Integrate absence/coverage workflow UI with REST API (F04)	Phúc	9	Implementation, debugging and peer review
5.4.1	Unit tests and coverage report	Minh	10	Add tests incrementally; do not wait for all coding to finish.
5.4.2	Automated cases (2 use cases × 2 scenarios), PR CI	Trí	10	Set up CI early; grow automated test coverage incrementally.
5.4.3	Test execution, defect fixes and PR reviews	Trí	12	Implementation, debugging and peer review
5.5.1	Deploy and verify live demo	Trí	6	Implementation, debugging and peer review
5.5.2	Demo video, implementation report, retrospective	Minh	6	Implementation, debugging and peer review
6.1.1	Prepare slides and project summary	Minh	4	Final date awaiting lecturer.
6.1.2	Demo script + practice all speakers	Trí	5	Each member presents >= 2 mins under rubric.
6.2.1	Final demo and presentation	Hoàng	2	Duration/effort provisional.
7.1.1	Weekly status meetings W02–W10	Hoàng	9	9 × 1 person-hour across team per meeting, adjust actual attendance.
7.1.2	Weekly Status Report W03	Phúc	2	Week 3 deadline must match the team-agreed reporting time; track actual evidence.
7.1.3	Weekly reports W02 and W04–W10 (8 reports)	Phúc	16	Create one issue per week (estimated 2h/report); do not backdate W02.
7.2.1	Maintain GitHub task board	Hoàng	4	Continuous planned total; split into weekly tasks.
7.2.2	Git/PR reviews, risks, baselines, change records	Hoàng	4	Continuous planned total.

4.4 Schedule

WBS / task	Owner	Planned finish	Status	% done	Notes
Preliminary Problem Statement	Phúc	19/10/2026	Done	100%	
Product Position Statement	Phúc	19/10/2026	Done	100%	
MoSCoW priorities, scope & exclusions	Phúc	19/10/2026	Done	100%	
Stakeholders and user interests	Minh	19/10/2026	Done	100%	
C4 System Context diagram	Minh	19/10/2026	Done	100%	
Technology stack and operating environment	Minh	19/10/2026	Done	100%	
Phase gates, exit criteria, roles	Hoàng	19/10/2026	Done	100%	
Three-level WBS for whole project	Hoàng	19/10/2026	Done	100%	
Estimate work-package person-hours	Hoàng	19/10/2026	Done	100%	
Gantt schedule, dependencies & critical path	Trí	19/10/2026	Done	100%	
Human resources and project cost	Trí	19/10/2026	Done	100%	
Risk register, RE, mitigation and owners	Trí	19/10/2026	Done	100%	
Repo, board, comms and access evidence	Minh	19/10/2026	Done	100%	
Final review, contribution, AI log, export	Hoàng	19/10/2026	Done	100%	



Minh chứng bằng Gantt Chart

4.5 Roles and Responsibilities

Role	Main	Backup	Responsibilities
Team lead / Project manager	Nguyễn Huy Hoàng	Lê Hoàng Phúc	Track deadlines, coordinate PAs and meetings, maintain project plan and baselines, integrate final documents, liaise with TA; support backend, testing and PR reviews.
Business analyst	Lê Hoàng Phúc	Võ Quang Minh	Prepare problem statement and scope, organize use cases and business rules, ensure requirements are understandable and reviewable; help with software development.
Designer / Architect	Võ Quang Minh	Nguyễn Huy Hoàng	Create C4/UML/data/UI designs, review consistency with SRS, contribute React/frontend and integration; backup verifies architecture decisions.
Developers	Lê Hoàng Phúc (backend/DB) ; Võ Quang Minh (frontend)	Nguyễn Huy Hoàng (backend); Nguyễn Cao Trí (integration)	All four members contribute to coding, integration, unit testing and pull request reviews. Development covers React frontend, Express REST APIs, Supabase authentication and authorization, constraint-based scheduling, roster publication and acknowledgement, and absence/shift coverage workflows.
Tester	Nguyễn Cao Trí	Lê Hoàng Phúc	Own test-plan execution, automated scenarios, defect reporting and regression testing; others provide peer testing, CI troubleshooting and code review.

4.6 Human Resources and Cost Estimate

The total effort estimate from table 4.3 is about 351 person-hours

Let assume a realistic junior-developer hourly rate of 90 000 VND/hour

Personnel cost: 351 × 90,000 = 31,590,000 VND.

For tool and infrastructure cost:

Github: free

Communication tools (outlock, microsoft team view,...): free

Domain host: free / about 200 000- 300 000 VND

AI tool assist (optional): free (student) / about 150 000 - 250 000 VND

Optional infrastructure and tool costs: 0–550,000 VND

=> Total estimated project cost: 31,590,000–32,140,000 VND.

5 Risk Management

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
1	Key member unavailable	People	3	4	12	Assign backup, critical module must be review early	Hoàng
2	Requirements change or remain ambiguous	Requirement	3	3	9	Freeze core business rules. Review rules carefully before coding starts	Hoàng
3	Task took longer than expected	Organisational	3	3	9	Setting a deadline earlier than necessary, to allow time for addressing issues	Hoàng
4	Frontend–backend integration problems (API contract mismatches, authentication header issues) appear late	Technology	3	4	12	Define and document API contracts early. Implement a thin integration layer and smoke-test it as soon as both sides have basic endpoints.	Minh
5	Supabase free-tier limits (connection count, storage, or rate limits) are hit during testing or the live demo	Technology	2	3	6	Monitor usage from the start. Prepare a fallback plan to switch to a local PostgreSQL instance or upgrade temporarily if needed.	Phúc

6 Tools & Collaboration Setup

- ✓ Moodle – announcements and PA submissions.
- ✓ Communication – MS Teams: General
- ✓ Git repository – GitHub: https://github.com/gloryh241106/ShiftManager. The instructor and TA have access:
  - main branch is protected; changes are merged through pull requests with at least one review
  - README (description, setup), .gitignore; never commit secrets (.env, API keys, passwords)
  - commit messages follow the Conventional Commits style (feat:, fix:, docs:, test:, …)
- ✓ Weekly status meeting (15 minutes) and weekly status report (template provided), committed to /docs/management/weekly every week from week 2 to week 10. The team chooses its weekly deadline and states it here.

Repository structure:

Folder	Content
/src	Source code
/docs/management	Project plan; weekly status reports in /docs/management/weekly
/docs/requirements	Vision, SRS, use cases, mockups
/docs/design	Architecture (C4), ADRs, UML models, UI design
/docs/test	Test plan, test cases, test reports
/docs/ai-usage	AI usage logs
/pa/PA1 … /pa/PA5	One sub-folder per PA submission (PDF + source files)

7 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Lê Hoàng Phúc	Gemini	Brainstorm and draft the Preliminary Problem Statement (2.1), Product Position Statement (2.2), and MoSCoW Scope & Key Features (2.4) for the ShiftManager project based on the PA0 specification.	The AI-generated draft was thoroughly reviewed against the course's sample specifications and the team's PA0 project scope.
2	Nguyễn Huy Hoàng	ChatGPT	Assisted with revising PA1 scope, WBS and preliminary estimates after TA feedback, including the proposal to replace two-way shift swapping with factory-team scheduling and absence coverage.	Pending human verification: Hoàng and Trí will check WBS IDs, F01–F04 scope, owners, person-hours and dependencies against PA0, PA1 rubric and Gantt; record the actual corrections before submission.
3	Võ Quang Minh	Claude (Anthropic)	Drafted Sections 2.3 (Stakeholders and Users), 3.1 (Preliminary Architecture – C4 System Context diagram), 3.2 (Technology Stack) and 3.3 (Hardware and Operating Environment), and added this log entry.	Pending human verification: Minh will check that stakeholders, the C4 diagram (users, external systems and relationships), technologies, hosting and hardware assumptions match the team's actual choices and PA0 scope, and record any corrections before submission.
4	Nguyễn Cao Trí	No AI tools were used		